Lab 9: Matrices for Models

What does a linear model assume about seedlings that share a plot?

Answer key. Answers are in magenta sans-serif type; longer answers have a bar in the margin.

The scenario

A restoration crew planted pinyon pine seedlings in three burned plots (A, B, and C) in a Great Basin woodland. A year later you measured each seedling's height growth (cm) and the soil moisture (%) next to it: 8 seedlings per plot, 24 in all. The six seedlings below are the first three in plots A and B.

Plot	Seedling	Soil moisture (%)
A	1	8
A	2	10
A	3	12
B	1	9
B	2	11
B	3	13

The crew's model is an ANCOVA: each plot has its own intercept, and all plots share one slope for soil moisture,

$$\text{growth}_i \sim \text{Normal}(\mu_i, \sigma^2), \qquad \mu_i = \beta_0 + \beta_B\,[\text{plot B}]_i + \beta_m\,\text{moisture}_i$$

where $[\text{plot B}]_i$ is 1 for a plot-B seedling and 0 otherwise. (For the six seedlings there are only two plots.)

How much does soil moisture matter, and what does this model assume about seedlings that grow side by side in the same plot?

Please keep laptops closed until the banner in Part 2 (except as noted in 1a). Part 1 and the group check-in are pen and paper.

Part 1: The model as matrices (individual, ~40 min)

1a. Write the model matrix X for the six seedlings, with columns for β_0, β_B, and β_m in that order. If you're stuck, you may open a laptop, read in the data (the code is in Part 2), and look at `model.matrix(~ plot + moisture, data = seedlings[c(1:3, 9:11), ])`. Then close the laptop again.

$$X = \begin{pmatrix} 1 & 0 & 8 \\ 1 & 0 & 10 \\ 1 & 0 & 12 \\ 1 & 1 & 9 \\ 1 & 1 & 11 \\ 1 & 1 & 13 \end{pmatrix}$$

1b. Take $\boldsymbol{\beta} = (2,\ 1.5,\ 0.5)^T$. Compute $X\boldsymbol{\beta}$, and say what each of the three numbers in $\boldsymbol{\beta}$ means for a seedling.

$X\beta = (6,\ 7,\ 8,\ 8,\ 9,\ 10)^T$ cm. $\beta_0 = 2$: expected growth of a plot-A seedling at 0% moisture (an extrapolation). $\beta_B = 1.5$: a plot-B seedling grows 1.5 cm more than a plot-A seedling at the same moisture. $\beta_m = 0.5$: each extra 1% of soil moisture adds 0.5 cm of growth, in either plot.

1c. Compute X^TX. What does each entry on its diagonal count or add up?

$$X^TX = \begin{pmatrix} 6 & 3 & 63 \\ 3 & 3 & 33 \\ 63 & 33 & 679 \end{pmatrix}$$

Diagonal: the number of seedlings (6), the number in plot B (3), and the sum of squared moisture values (679).

1d. A reminder about diagonal matrices (all off-diagonal entries 0): the determinant is the product of the diagonal entries, and the inverse is also diagonal, with each diagonal entry replaced by its reciprocal. For example,

$$\begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/5 \end{pmatrix}, \qquad \begin{vmatrix} 2 & 0 \\ 0 & 5 \end{vmatrix} = 2 \times 5 = 10.$$

Suppose the six growth values are independent with $\sigma = 0.5$ cm. Write the covariance matrix V, its inverse, and its determinant. Then suppose plot B is more variable, with $\sigma_B = 1$ cm (plot A stays at 0.5). Write the new V, and write the quadratic form $\mathbf{r}^TV^{-1}\mathbf{r}$ for a residual vector $\mathbf{r} = (r_1, \ldots, r_6)^T$ as a sum. How much does a plot-B residual contribute to $\mathbf{r}^TV^{-1}\mathbf{r}$, compared with a plot-A residual of the same size?

$V = 0.25\,I$ (6×6), $V^{-1} = 4\,I$, and $|V| = 0.25^6 = 1/4096 \approx 2.44 \times 10^{-4}$.

With $\sigma_B = 1$: $V = \mathrm{diag}(0.25, 0.25, 0.25, 1, 1, 1)$ and

$$\mathbf{r}^TV^{-1}\mathbf{r} = 4(r_1^2 + r_2^2 + r_3^2) + (r_4^2 + r_5^2 + r_6^2)$$

A plot-B residual contributes one quarter as much as a plot-A residual of the same size (r^2 vs. $4r^2$).

1e. Two seedlings grow side by side in the same plot, and their residuals have correlation 0.5 (each with variance 1):

$$V = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$$

Find $|V|$ and V^{-1}. Compute $\mathbf{r}^TV^{-1}\mathbf{r}$ for $\mathbf{r} = (1, 1)^T$ and for $\mathbf{r} = (1, -1)^T$, and compare both with independence ($V = I$). Which pattern of residuals is more surprising when the seedlings are correlated, and why does that make sense?

$|V| = 1 - 0.25 = 0.75$, and $V^{-1} = \dfrac{1}{0.75}\begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} = \begin{pmatrix} 4/3 & -2/3 \\ -2/3 & 4/3 \end{pmatrix}$.

$(1, 1)$: 1.33. $(1, -1)$: 4.00. Under independence both are 2.

Residuals of opposite sign are more surprising. Correlated seedlings are expected to be high or low together, so two seedlings both growing more than predicted is less unusual than one high and one low.

1f. Three seedlings, three proposed correlations:

$$C = \begin{pmatrix} 1 & 0.9 & -0.9 \\ 0.9 & 1 & 0.9 \\ -0.9 & 0.9 & 1 \end{pmatrix}$$

Could this be the correlation matrix of their growth? Explain without computing anything.

> No. Seedlings 1 and 2 rise and fall together, and so do 2 and 3, so 1 and 3 must also be positively correlated; they can't be strongly negatively correlated.

Group check-in (~15 min)

Compare your answers with your group, especially 1c, 1e, and 1f, and revise as needed. Before you start coding, agree on what the rows and columns of X stand for, and what the diagonal and off-diagonal entries of V say about the seedlings. For 1d and 1e, check that you agree on how changing V changes how much each residual counts in $\mathbf{r}^T V^{-1} \mathbf{r}$.

Laptops open now.

Part 2: Check it with R (laptop, ~50 min)

Using AI: each question has a recommendation. *Recommended: no AI* marks the core skill of the lab (building and using model and covariance matrices); R's help pages, web search, and classmates are fine. *AI optional* marks work where the results matter but writing the code by hand isn't the point. After each question, circle how you actually used AI.

What to turn in: your filled-in answers. You don't need to submit your code.

The data for all 24 seedlings are on the course website:

```
data_url <-
  "https://kevintshoemaker.github.io/NRES-746/data/PinyonSeedlings.csv"
seedlings <- read.csv(data_url)
```

2a. *(Recommended: no AI.)* Build X for all 24 seedlings with `model.matrix(~ plot + moisture, data = seedlings)`. Solve the normal equations, $X^T X\hat{\boldsymbol{\beta}} = X^T \mathbf{y}$, using `%*%`, `t()`, and `solve()`, and check your answer against `lm()`. Check: the moisture slope should be 0.44. Fill in the table.

Coefficient	Normal equations	lm()
(Intercept)	*3.03*	*3.03*
plotB	*1.63*	*1.63*
plotC	*−0.77*	*−0.77*
moisture	*0.44*	*0.44*

AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

2b. *(Recommended: no AI.)* Let $\mathbf{r} = \mathbf{y} - X\hat{\boldsymbol{\beta}}$ be the vector of residuals at your $\hat{\boldsymbol{\beta}}$, with r_i the residual for seedling i. At $\hat{\boldsymbol{\beta}}$ and the MLE $\hat{\sigma} = \sqrt{\sum r_i^2/n}$ (with $n = 24$), compute the log-likelihood two ways: with `sum(dnorm(..., log = TRUE))`, and with `mvtnorm::dmvnorm(..., log = TRUE)` using $V = \hat{\sigma}^2 I$.

$\hat{\sigma} =$ 0.87 cm. `sum(dnorm())`: -30.70 `dmvnorm()`: -30.70

AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

2c. *(AI optional.)* Check your answers to 1e and 1f with `det()`, `solve()`, and `eigen()`. Then try to draw 1000 random values for three seedlings from a multivariate normal with correlation matrix C from 1f, using `MASS::mvrnorm()`. What happens?

> `mvrnorm()` stops with an error ("'Sigma' is not positive definite").

A reminder about eigenvalues: for a covariance or correlation matrix, each eigenvector picks out one weighted combination of the variables (here, of the three seedlings' growth), and its eigenvalue is the variance of that combination. A variance can't be negative, so every eigenvalue of a valid covariance or correlation matrix must be zero or positive.

Run `eigen(C)` and fill in the eigenvalues. Which one breaks the rule? Its eigenvector (the matching column of `eigen(C)$vectors`) is proportional to $(1, -1, 1)$, the combination $G_1 - G_2 + G_3$, where G_j is seedling j's growth on a standardized scale (variance 1). Use the rule for the variance of a sum and the correlations in C to compute $\mathrm{Var}(G_1 - G_2 + G_3)$ by hand. What does your answer say about whether C could describe real seedlings?

Eigenvalues of C: 1.9, 1.9, -0.8

> The third, -0.8, is negative.
>
> $$\mathrm{Var}(G_1 - G_2 + G_3) = 1 + 1 + 1 + 2\,[(-1)(0.9) + (1)(-0.9) + (-1)(0.9)] = 3 - 5.4 = -2.4.$$
>
> A negative variance is impossible, so no three seedlings could have these correlations: C is not a valid correlation matrix (not positive definite), which is why `mvrnorm()` refuses it.

AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

2d. Where did you use AI, if anywhere, and what did you have to check or fix? If you didn't use it, what was hardest to get working?

Open-ended.

Group discussion and revision (~15 min)

Close laptops. Compare your Part 2 results with your group, especially anything that didn't match in 2a or 2b and what the eigenvalues in 2c told you, and revise as needed. Record any changes you made to earlier answers, new insights, and any "muddiest points" or topics you want to review.

Open-ended.

Bonus (optional). Here is a different covariance matrix someone proposed for the six seedlings in the table (rows and columns in the same order: A1, A2, A3, B1, B2, B3):

$$V = \begin{pmatrix} 1 & 0.6 & 0.6 & 0 & 0 & 0 \\ 0.6 & 1 & 0.6 & 0 & 0 & 0 \\ 0.6 & 0.6 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0.6 & 0.6 \\ 0 & 0 & 0 & 0.6 & 1 & 0.6 \\ 0 & 0 & 0 & 0.6 & 0.6 & 1 \end{pmatrix}$$

What does this V claim about the seedlings, and what could make the 0.6s biologically reasonable? Should the crew treat its 24 seedlings as 24 independent replicates when reporting the effect of plot?

> Each seedling has variance 1; seedlings in the same plot are correlated (0.6) and seedlings in different plots are independent. Shared soil, burn severity, microclimate, herbivores, or planting date could make seedlings in a plot grow alike. If so, the 24 seedlings aren't 24 independent replicates of "plot": plot differences rest on three plots, and treating seedlings as replicates overstates the evidence.